

Optimal Approach:

- Source Declaration (Easier referencing, ensure linearity)
- Data Transformation:
 - Retrieve data from source, conduct small alterations to tidy up raw data
- Modular Transformation:
 - Cleaning up data and merging necessary tables to ensure modularity and readability
- Query from the modular tables

Setup Instructions:

1. Open the dataset in Google BigQuery
2. Go to IAM, ensure the service account has the following roles:
 - a. BigQuery Admin
 - b. BigQuery Data Editor
 - c. BigQuery Data Viewer
 - d. BigQuery Job User
 - e. BigQuery User
3. Ensure *workflow_settings.yaml* has the following settings:

`defaultProject: bigquery-public-data`

`defaultLocation: US`

`defaultDataset: ncaa_basketball`

`defaultAssertionDataset: dataform_assertions`

`dataformCoreVersion: 3.0.0`

4. Declare source
5. Conduct data and modular transformation
6. Query from the transformed tables

Star Schema:

Fact Table:

mbb_pbp_sr

Dimension Table:

mbb_players_games_sr

mbb_teams

mbb_teams_games_sr

mbb_historical_teams_seasons



mbb_pbp_sr is chosen as fact table because it contains the most foreign keys to different primary keys.

Directory Structure Explanation:

source_declarations folder: Contains source declarations files

staging folder : Contains data transformed files (supposedly to include modular transformed files, and then to create star schema tables from there)

Queries folder : Contains query files(SQL) for the 3 selected questions (q1-3)

Tables folder : Contains output tables from queries

README.pdf : Contains explanations, approaches and setup instructions